

FS-2603 · Obligation-Safe Automated Investing

An investment engine that never misses a bill.

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Idle cash is a tax on uncertainty.

Urban salaried Indian households keep months of cash idle because automating investments risks missing a fixed auto-debit. Income lands across a 6-day window. Rent and EMIs don't.

₹2L

Idle for 3 years

Forfeits 7–9% real return annually

6d

Income window

vs. fixed obligation due dates

0

Tolerance for misses

One missed obligation costs more than automation gains

Source: RBI Household Financial Savings FY24

One path, deterministic, single-writer.



State Ownership

Immutable history vs mutable ledger, single-writer scored path

Consistency

Strong, single-node, log-structured

Failure Path

Forecast fails → invest 0 ·
Settlement rejects → retry ·
Market halt → freeze ·
Insufficient cash → block



The forecast and the exit constrain each other.

The forecast determines how much can be invested. Settlement lag $T+2$ means the decision is made cycles before the obligation falls due.

A confident forecast still breaches if the exit lags. We trade upside for the guarantee of zero missed obligations, because the metric gates on zero misses.

Every number is measured, not claimed.

Forecast Latency

<3s on 1 CPU, 1 GB

P90 Coverage

85–95% target

Missed Obligations

0

Reconciliation

Exact to the paisa

Compute Footprint

Sampled with psutil

Cost / 1,000 Rebalances

Compute × cloud rate

Two rejected alternatives.

Rejected A — Deep Learning Forecast

LSTM / Transformer

- Non-deterministic
- Exceeds compute budget
- Needs training data we don't have

Rejected B — Multi-Node Consensus Ledger

Distributed architecture

- Adds overhead
- Zero scoring benefit on 1 CPU

❏ Both rejections preserve the core constraint: obligation safety on minimal compute.

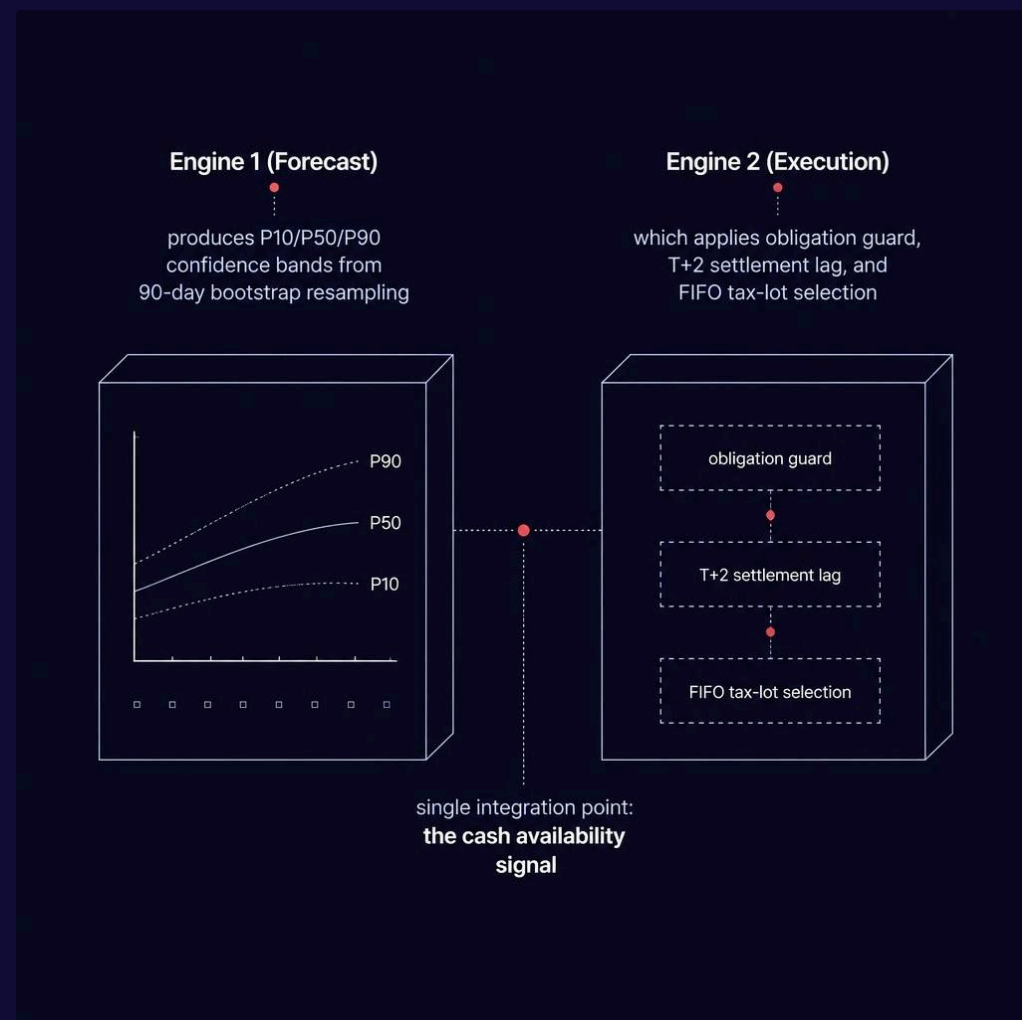
Forecast and Execution, one integration point.

Engine 1 — Forecast

Bootstrap resampling over 90-day history producing P10 / P50 / P90 over 60 days.

Engine 2 — Execution

Obligation guard rejects any trade pushing P5 cash below the next obligation. Settlement lag T+2 modeled. FIFO tax-lot selection explicit and auditable.



Config, not rewrite.

Income drops to zero on day 12. Market falls 18% in 4 minutes. One holding halts.

→ **Regime-break detector**

Flags distribution shift; intervals widen automatically

→ **Halted position**

Frozen in place; no forced liquidation

→ **All thresholds in `config.toml`**

Absorbed with a config edit, not code

Every claim reproducible from the repo.

01

Calibration

Reliability diagram

02

Reconciliation

diff = 0

03

Determinism

Double-run hash diff

04

Compute

psutil samples

05

Cost

Compute-time × cloud rate

Three honest failure conditions.

1

Bootstrap assumes stationarity

Underestimates downside when income regime breaks.

Mitigation: regime-break detector

2

Settlement model breaks on unannounced partial fills

Mitigation: reconcile each cycle, re-reserve on mismatch

3

Obligation guard fails on late-registered obligations

Mitigation: forward-declared obligations only, refuse trades on unregistered windows

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